Cholestasis of Prematurity and Parenteral Nutrition: Sepsis, Necrotizing Enterocolitis, and the Hidden Biliary Atresia.

Picture a baby born at twenty-seven weeks of gestation, now six weeks old, in the neonatal intensive care unit. She had necrotizing enterocolitis, the inflammatory bowel necrosis of the premature gut usually shortened to NEC, at two weeks of age. She was treated medically, had two episodes of bloodstream infection from her central line, and is still receiving most of her calories by vein. Her direct bilirubin, the conjugated fraction that the liver has processed but cannot excrete, has climbed from zero point eight to three and a half milligrams per deciliter. Everyone on the team has the same explanation ready: it is the parenteral nutrition. And most of the time, they will be right.

Now picture a second baby, born at thirty weeks, who has been on full enteral feeds for two weeks. His parenteral nutrition was stopped ten days ago. His direct bilirubin is five and still rising, and the nurse mentions that his stools have looked pale for a few days. The chart still says "parenteral-nutrition-associated cholestasis," because that label was written three weeks ago and nobody has taken it off.

The clinical question of this chapter is how to manage cholestasis in the preterm infant who has been on parenteral nutrition, and how to recognize the few infants in whom that label is hiding something else, most importantly biliary atresia. The main idea is simple to state. Cholestasis in a preterm infant on parenteral nutrition is usually multifactorial and reversible, but parenteral-nutrition-associated cholestasis is a diagnosis of exclusion. It must follow the dose of parenteral nutrition, it must improve as feeds advance, and it must never be accepted in an infant with pale stools. The stakes are real. In a national cohort from Taiwan, preterm infants with biliary atresia had their Kasai operation at a mean of seventy-two days of age, against fifty-three days in term infants, and fewer of them kept their own liver.

We will walk through the topic in order: definitions and how common the problem is, the risk factors, the mechanism, the special role of sepsis and NEC, the surgical infant and the progression to intestinal failure liver disease, the lipid emulsion evidence, the other treatments, and then the biliary atresia problem, the guidelines, the traps, and what to do differently on the ward.

A note on sources. For the cholestatic infant in general we use the 2017 joint guideline on cholestatic jaundice in infants by Fawaz and colleagues for the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN. For the epidemiology we use the Intermountain cohort reported by Christensen and colleagues in 2007, the meta-analysis by Lauriti and colleagues in 2014, the Korean cohort by Kim and colleagues in 2016, and the Children's Hospital Neonatal Database analysis of NEC by Nayak and colleagues in 2024. For the mechanism we use the mouse studies by El Kasmi and colleagues from Colorado. For lipid emulsions we use the Boston fish oil series by Puder and colleagues, the United States label for fish oil emulsion, a randomized trial by Repa and colleagues in Vienna, a Cochrane

review by Kapoor and colleagues in 2019, and the 2018 pediatric parenteral nutrition guidelines written jointly by ESPGHAN and the adult, perinatal and Chinese nutrition societies. For biliary atresia in preterm infants we use the Taiwanese cohort by Chiu and colleagues in 2013 and the Dutch national cohort by van Wessel and colleagues in 2017.

Part one. Definitions and how common it is.

The first thing to notice is that different thresholds serve different purposes. The 2017 joint guideline says that a direct or conjugated bilirubin above one milligram per deciliter is abnormal in any infant, regardless of the total bilirubin. That is the threshold for starting an evaluation. The research literature, and the United States label for fish oil emulsion, define parenteral-nutrition-associated cholestasis, which we will call PNAC, more strictly: a direct bilirubin of two milligrams per deciliter or more, usually after at least fourteen days of parenteral nutrition. Some older trials used a third definition, a direct fraction above fifteen percent of the total bilirubin. That ratio definition inflates the rates dramatically. In one trial that used it, sixty-nine percent of infants qualified as cholestatic. So before comparing any two studies, check the definition. At the bedside, investigate at one milligram per deciliter, and reserve the label of PNAC for an infant who meets two milligrams per deciliter on prolonged parenteral nutrition and whose course fits.

How common is PNAC? The most useful single picture comes from the Intermountain Healthcare neonatal units in Utah, reported by Christensen and colleagues. They followed nine thousand eight hundred and sixty-one neonates. Six thousand five hundred and forty-three received parenteral nutrition, one thousand three hundred and sixty-six received it for at least fourteen days, and three hundred and fifty-seven developed liver disease. The key finding was a dose response. Among infants who received parenteral nutrition for two to four weeks, fourteen percent became cholestatic. For four to eight weeks, forty-three percent. For eight weeks to a little over three months, seventy-two percent. And beyond one hundred days, eighty-five percent. The bar chart rises steadily, almost in a straight line. The risk was also strongly tied to size and to the underlying gut problem. Compared with larger infants, a birth weight below five hundred grams carried an odds ratio of about thirty-one, and five hundred to seven hundred and fifty grams an odds ratio of about thirteen. Jejunal atresia carried an odds ratio of twenty-four and gastroschisis an odds ratio of about twenty. In other words, the infants at highest risk were the smallest and those whose intestine could not be used.

The pooled data say the same thing. A meta-analysis by Lauriti and colleagues of twenty-three studies and three thousand two hundred and eighty infants found PNAC in about twenty-eight percent overall. With parenteral nutrition for a month or less, the rate was about sixteen percent. With two months or more, it was about sixty-one percent. In infants with intestinal failure, liver disease was present in about half. An older study by Beale and colleagues from 1979 described cholestasis in about half of infants under one thousand grams after two to four weeks of parenteral nutrition, and in about fifteen percent of those between one and two thousand grams.

It is worth noting that the 2017 joint guideline describes parenteral nutrition cholestasis as occurring in up to one fifth of neonates who receive parenteral nutrition for more than two weeks. That figure is fair for a short course, but it understates the risk for the infant who has been on parenteral nutrition for two months, which is exactly the infant you are worried about.

Part two. What adds to the risk.

Parenteral nutrition is rarely the only factor. In a Korean cohort of very low birth weight infants reported by Kim and colleagues, forty-six of one hundred and eighty-seven infants developed PNAC. In the adjusted model, the strongest independent predictor was the duration of antibiotic treatment, with an odds ratio of about four and a half. A long period without enteral feeds came next, with an odds ratio of about two point nine, then the duration of parenteral nutrition at about two point seven, NEC at about two point four, and a gastrointestinal operation at about two point two. Sepsis was far more common in the cholestatic infants, twenty-eight percent against four percent, but it was not independent once antibiotic days were in the model, which simply means that antibiotic days were capturing the burden of infection. In that cohort cholestasis began at a mean of about fifty-two days of life, the direct bilirubin peaked at about seven milligrams per deciliter around day seventy, and the cholestasis lasted about two months.

Being small for gestational age adds risk on its own. Two separate cohorts, by Lee and colleagues and by Robinson and Ehrenkranz, found that small-for-gestational-age infants had about three times the odds of PNAC. In the Lee study severe PNAC occurred in sixty-one percent of small-for-gestational-age infants against thirty-five percent of appropriately grown infants.

Part three. The mechanism: five hits on a small bile acid pool.

It helps to think of the mechanism as a series of hits that converge on the canalicular transporters, the pumps in the hepatocyte membrane that push bile acids and other molecules into bile.

The first hit is the absence of enteral feeding. When nothing enters the gut, bile acids are not reabsorbed in the ileum, so the intestinal hormone fibroblast growth factor nineteen, which normally signals back to the liver, stays low. Cholecystokinin is not released, the gallbladder does not empty, and bile stagnates. The preterm infant also starts with a small and immature bile acid pool.

The second hit comes from the lipid emulsion itself. Traditional soybean oil emulsions contain plant sterols called phytosterols. One of them, stigmasterol, acts as an antagonist of the farnesoid X receptor, usually shortened to FXR, which is the nuclear receptor that senses bile acids and switches on the transporters that export them. This was shown by El Kasmi and colleagues in a mouse model published in 2013.

The third hit follows from the second. With FXR signaling blocked, the bile salt export pump, often called BSEP, the canalicular conjugate pump called MRP2, and the sterol exporters fall. Bile acids and phytosterols are retained in the hepatocyte, which compounds the injury.

The fourth hit comes from the gut. An injured, unfed intestine leaks bacterial endotoxin, or lipopolysaccharide, into the portal blood. The resident macrophages of the liver, the Kupffer cells, respond through toll-like receptor four and release inflammatory cytokines such as interleukin six and tumor necrosis factor.

The fifth hit is sepsis. Those same cytokines, and endotoxin itself, switch off the transcription of the sodium taurocholate cotransporting polypeptide, which takes bile acids up from the blood, and of the bile salt export pump. This is the cholestasis of sepsis, and it can appear abruptly in an infant who was previously stable.

The experimental evidence for this two-hit model is quite striking. In the 2012 mouse study by El Kasmi and colleagues, neither soybean lipid nor gut injury alone was enough. When mice with chemically induced colitis were given soybean-based parenteral nutrition, serum bile acids rose more than tenfold and liver enzymes about fourfold, and the Kupffer cells expressed about forty times more transforming growth factor beta. When the gut bacteria were suppressed with antibiotics, or when the mice carried a nonfunctional toll-like receptor four, the liver was protected. In older work by Trauner and colleagues, endotoxin cut the binding of the main transcription factor at the uptake transporter promoter to about forty-four percent of baseline, with the lowest messenger RNA levels at sixteen hours. And in neonatal pigs on parenteral nutrition, Jain and colleagues showed that giving the bile acid chenodeoxycholic acid by mouth lowered the direct bilirubin almost fourfold and restored fibroblast growth factor nineteen. Taken together, these studies describe exactly the infant we care about: a preterm baby with NEC or a surgical gut, on prolonged soybean-based parenteral nutrition, with recurrent infection.

The model also explains why fish oil helps. Fish oil emulsions contain almost no phytosterols, more of the antioxidant alpha tocopherol, and less omega-six fatty acid. In the mouse experiments, fish oil parenteral nutrition prevented the injury that soybean lipid caused.

Part four. Sepsis and necrotizing enterocolitis.

The largest recent picture of NEC and cholestasis comes from the Children's Hospital Neonatal Database, analyzed by Nayak and colleagues in 2024, covering one thousand four hundred and seventy-two infants with NEC. Cholestasis developed in about thirty-eight percent of infants with medical NEC and in about seventy percent of those with surgical NEC, almost double. Among infants with medical NEC, a bloodstream infection within two days of the NEC diagnosis nearly tripled the odds of cholestasis, with an odds ratio of about two point eight, and each additional day of parenteral nutrition added about eight percent to the odds. Cholestasis also marked the sicker infant: in medical NEC, mortality was about ten percent among infants with cholestasis and two and a half percent among those without. A separate cohort by Garg and colleagues found cholestasis independently associated with NEC-associated sepsis, with an odds ratio of about three. A Chinese multicenter cohort found that surgical NEC carried an odds ratio of about eleven, while a mixed lipid emulsion and breast milk were each protective, roughly halving the risk or better.

The practical point is that a new rise in direct bilirubin in a stable infant on parenteral nutrition is a reason to look for infection. The bilirubin often follows the infection, rising with it and falling after

it is treated.

Part five. The surgical infant and intestinal failure liver disease.

When cholestasis persists for months in an infant with a short or nonfunctioning intestine, it becomes part of what is called intestinal-failure-associated liver disease. The infants at risk are those with gastroschisis, jejunal atresia, surgical NEC with short bowel, and high-output stomas. They carry additional hits: bacterial overgrowth and translocation, recurrent central line infections, and months rather than weeks of parenteral nutrition. The disease can progress to fibrosis, portal hypertension and cirrhosis. The 2015 ESPGHAN working group position paper, led by Lacaille, described liver disease in about a quarter of children on home parenteral nutrition and mortality of up to forty percent once the disease is established, and recommended early enteral feeding and early referral to an intestinal failure center.

Two points matter for follow-up. First, fibrosis lags behind bilirubin. The bilirubin can normalize on fish oil while fibrosis persists, so the infant should be followed with platelet counts, spleen size and, where available, elastography, not with bilirubin alone. Second, reversal is possible even late. In a Boston series by Nandivada and colleagues of fifty-one children with parenteral-nutrition-associated cirrhosis, cholestasis resolved on fish oil in thirty-nine of them, at a median of seventy-four days, and none of those thirty-nine needed a transplant.

Part six. The lipid emulsion: the first lever.

The fish oil story began in Boston. Puder and colleagues compared forty-two infants with established cholestasis who received a pure fish oil emulsion at one gram per kilogram per day with forty-nine historical controls who had received soybean oil. Cholestasis reversed while still on parenteral nutrition in nineteen of thirty-eight infants on fish oil, about half, against two of thirty-six controls, about six percent. Three of forty-two fish oil infants died, about seven percent, against twelve of forty-nine controls, about a quarter. One fish oil infant was transplanted against six controls. Overall, cholestasis reversed about six times faster on fish oil. There are two important caveats. The controls were historical, so other aspects of care had changed between eras. And one gram per kilogram per day of fish oil is also a reduction in total lipid, so lipid restriction and lipid type are both part of the effect.

The United States Food and Drug Administration approved a pure fish oil emulsion, Omegaven, in 2018 for children with PNAC. The licensed dose is one gram per kilogram per day, which is also the maximum. In the efficacy analysis, eighty-two treated infants were compared with forty-one historical soybean controls, and the median time to a direct bilirubin below two milligrams per deciliter was about five to six weeks. In the safety population, sixty percent reached a direct bilirubin below two with transaminases below three times the upper limit of normal, but mortality was thirteen percent and five percent were transplanted, a reminder that these are sick infants. While on fish oil, watch for essential fatty acid deficiency using the triene to tetraene ratio, for growth on a lower lipid energy load, and for a bleeding tendency in an infant who may already be coagulopathic.

Prevention has been harder to prove than rescue. In a double-blind randomized trial in Vienna, Repa and colleagues randomized two hundred and twenty-three extremely low birth weight infants to a mixed lipid emulsion containing soybean, medium-chain triglycerides, olive oil and fish oil, known as SMOF lipid, or to soybean oil. PNAC occurred in about ten percent on the mixed emulsion against about sixteen percent on soybean, a difference that did not reach statistical significance. The 2019 Cochrane review by Kapoor and colleagues, covering twenty-nine trials and about two thousand preterm infants, found a relative risk of about zero point six for cholestasis with fish-oil-containing emulsions, but with a confidence interval that crossed one and certainty rated low. A more recent meta-analysis of twenty randomized trials in neonates, by He and colleagues in 2024, found that the mixed emulsion reduced cholestasis by about a third, with no heterogeneity between trials. So the honest summary is that composite lipid probably lowers PNAC modestly.

What about simply giving less soybean lipid? Levit and colleagues randomized infants born at twenty-nine weeks or less to about one gram versus about three grams per kilogram per day of soybean emulsion. Cholestasis rates were sixty-nine percent and sixty-three percent, with no significant difference, using the generous ratio definition. Restriction of soybean lipid alone does not prevent PNAC, and it costs energy for growth. Change the lipid, and above all feed the gut.

Part seven. The other levers.

Enteral feeding is the most reliable treatment and the most reliable prevention. Trophic feeds should start early and advance as tolerated, and breast milk appears protective. Everything else is secondary.

Ursodeoxycholic acid, the hydrophilic bile acid often shortened to UDCA, is widely used. In a small cohort of very low birth weight infants, Chen and colleagues found that cholestasis lasted about sixty-three days in treated infants against about ninety-two days in controls. But there is no definitive randomized trial, and a Cochrane protocol noted that UDCA use in American neonatal units rose by eighty-six percent between 1999 and 2004 without one. It also needs an enteral route to work.

Erythromycin, used as a prokinetic, helps feeding intolerance at high doses of forty to fifty milligrams per kilogram per day, and the 2022 Cochrane review found benefit only in infants older than thirty-two weeks, with no reported effect on cholestasis. Within the parenteral nutrition prescription itself, avoid excess glucose and calories, cycle the infusion when feasible, and remember that copper and manganese are excreted in bile, so they should be reviewed in prolonged cholestasis. Cycling has physiological logic, but there is no verified randomized trial in preterm infants. Use these measures, but do not let them substitute for feeds.

Part eight. The hidden biliary atresia.

Biliary atresia is not a disease of term infants only. In the Taiwanese national cohort reported by Chiu and colleagues, which used the stool color card screening program, the incidence of biliary atresia was about two point four per ten thousand live births in preterm infants against about one

point four in term infants. The preterm infants did worse on every outcome. Only about forty-four percent had their Kasai before sixty days, against about sixty-nine percent of term infants. Only thirty-seven percent were jaundice-free at three months, against sixty-two percent. And native liver survival at eighteen months was fifty percent against about seventy-three percent. The mean age at Kasai was about seventy-two days against fifty-three. Importantly, the stool card was just as sensitive in preterm infants, about ninety-six percent before sixty days. The card works if it is used.

The Dutch national cohort reported by van Wessel and colleagues tells a similar story. The incidence of biliary atresia in preterm infants was about one per ten thousand, roughly double that of the whole population, the median Kasai was at seventy days, and native liver survival at four years was twenty-one percent against forty-six percent for all infants with biliary atresia. Not every cohort agrees on outcome. At King's College Hospital in London, preterm infants made up about three percent of six hundred and ninety-two infants with biliary atresia, and their Kasai was performed at a median of sixty-five days against fifty-six in matched term controls, but bile drainage was similar. A Rome series also found similar clearance rates. In the Japanese registry, infants with a birth weight under one thousand five hundred grams made up less than one percent of all cases. The consistent signal across these studies is delay. The preterm infant is already on a ward, already jaundiced, and already explained.

Part nine. Why every test is confounded.

Each test we use for biliary atresia behaves differently in a preterm infant on parenteral nutrition. Stool color is the most important clue, but an unfed infant passes little stool, so color is hard to judge. Gamma-glutamyl transferase, the canalicular enzyme usually shortened to GGT, is high in both PNAC and biliary atresia and does not separate them. The hepatobiliary scintigraphy scan, often called a HIDA scan, has a sensitivity of about ninety-nine percent but a specificity of only about seventy percent in the meta-analysis cited by the 2017 guideline, and the guideline notes that parenteral nutrition and low birth weight themselves cause non-excretion. On ultrasound, the triangular cord sign has a sensitivity of about seventy-four percent and a specificity of about ninety-seven percent, and gallbladder abnormalities a sensitivity of about eighty-five percent and specificity of about ninety-two percent, but a fasting gallbladder in an infant on parenteral nutrition can look small or absent. Liver biopsy performs best, with a sensitivity of one hundred percent and specificity of about ninety-four percent in the guideline's summary, yet parenteral nutrition cholestasis can mimic obstruction under the microscope. Finally, newborn direct bilirubin screening, studied in Texas by Harpavat and colleagues in more than one hundred and twenty thousand infants, had a sensitivity of one hundred percent and specificity of about ninety-nine point nine percent, but preterm infants screened positive at the first step about three times as often as term infants, and the program was not designed around the neonatal unit.

So the practical approach has five steps. First, screen: measure the direct bilirubin at about two weeks of parenteral nutrition and every one to two weeks after that, and act when it exceeds one milligram per deciliter. Second, look: check stool color at every diaper change, and let pale stools override every other explanation. Ontario's newborn screening program, for example, keeps preterm infants on stool card screening in the neonatal unit until a month past their due date. Third,

exclude the treatable causes: sepsis, urinary tract infection, hypothyroidism, cytomegalovirus, metabolic disease and drugs. Fourth, image: do an ultrasound after a feed if possible, looking at the gallbladder, the triangular cord and the spleen. Fifth, escalate: if the bilirubin keeps rising despite feeds, the stools are pale, or the scan is abnormal, proceed to biopsy and a cholangiogram with the surgeons.

The single most useful test is the trajectory. PNAC follows the parenteral nutrition: it plateaus and then falls as feeds reach full volume. A direct bilirubin that keeps rising after parenteral nutrition stops, or stools that stay pale on full feeds, means obstruction until proven otherwise.

Part ten. What the guidelines say.

The guidelines agree on when to investigate. The 2017 joint North American and European guideline sets the threshold at a direct bilirubin above one milligram per deciliter, and that applies on both sides of the Atlantic.

They differ on pure fish oil. In the United States, the Food and Drug Administration licenses pure fish oil at one gram per kilogram per day for established PNAC. The 2018 European pediatric parenteral nutrition guidelines treat pure fish oil as a short-term rescue and recommend composite emulsions containing fish oil as the first-line lipid for children on long-term parenteral nutrition with cholestasis. They also differ on the default lipid. The European guideline recommends replacing pure soybean emulsion with a composite emulsion for any parenteral nutrition longer than a few days. In the United States, the mixed emulsion was licensed for neonates in 2022, but the 2023 preterm parenteral nutrition guideline from the American Society for Parenteral and Enteral Nutrition rated the evidence as very low certainty and did not make a firm recommendation on liver outcomes.

There is a material gap in how common the problem is said to be. The joint guideline's figure of up to one fifth is much lower than the cohort data, which show cholestasis in forty-three percent after four to eight weeks of parenteral nutrition and in seventy-two to eighty-five percent beyond eight weeks. And there are true gaps. No society makes a firm recommendation about ursodeoxycholic acid, cycling or erythromycin for PNAC. No guideline describes how to screen preterm infants for biliary atresia in the neonatal unit. And neither the American nor the European liver societies have specific guidance on PNAC.

Part eleven. Traps and pearls.

There are four common traps. The first is saying it is the parenteral nutrition without a trajectory to support it; with pale stools, think biliary atresia and ask for the stool card. The second is treating a non-excreting HIDA scan as proof of atresia; non-excretion is expected on parenteral nutrition and in low birth weight infants, and the specificity is only about seventy percent. The third is cutting soybean lipid to one gram per kilogram per day and expecting prevention; restriction alone did not prevent PNAC in the randomized trial, and it costs growth. Change the lipid and feed the gut. The fourth is relaxing because the bilirubin is now normal; fibrosis can persist, so follow platelets and spleen size, and watch for essential fatty acid deficiency on fish oil.

The numbers worth carrying into the neonatal unit are these. PNAC is dose dependent: fourteen percent after two to four weeks of parenteral nutrition, forty-three percent after four to eight weeks, and seventy-two to eighty-five percent beyond that. Antibiotic days, a lack of enteral feeds, parenteral nutrition days, NEC and surgery each add risk, and being small for gestational age about triples it. Surgical NEC brings cholestasis in seventy percent against thirty-eight percent in medical NEC. Phytosterols antagonize FXR, gut endotoxin activates Kupffer cells, and sepsis switches off the uptake and export pumps. Fish oil at one gram per kilogram per day rescues established PNAC over a median of about six weeks, while composite lipid modestly prevents it. Preterm infants with biliary atresia have their Kasai later, at about seventy-two days rather than fifty-three, and keep their native liver less often, but the stool card still works.

What to do differently in clinic.

When you meet a preterm infant with a rising direct bilirubin, start by writing down the trajectory: how many days of parenteral nutrition, what volume of feeds, and what the bilirubin has done over the last two weeks. Look at the stools yourself, and ask the nurses for the stool card. If the infant is unwell, or the bilirubin has jumped, look for infection before you blame the nutrition. Move the infant to a composite lipid emulsion containing fish oil if parenteral nutrition is going to continue, push enteral feeds as far as the gut allows, and keep pure fish oil at one gram per kilogram per day for established, progressive cholestasis, watching growth and fatty acids. For the infant with a surgical gut and months of parenteral nutrition, involve an intestinal failure team early and follow platelets and spleen size, not only bilirubin. And above all, take the label off the chart when it no longer fits. If the bilirubin keeps rising after the parenteral nutrition has stopped, or the stools are pale on full feeds, treat it as biliary atresia until a cholangiogram says otherwise, because for that baby every week counts.

End of chapter.